Yuan Chen

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Research Interests

Astrochemistry, (sub)millimeter & infrared observations (ALMA & JWST), interstellar medium, complex organic molecules (COMs), star-forming regions, protostars (hot cores/corinos)

Employment

Max Planck Institute for Extraterrestrial Physics (MPE) — Garching, Germany
Postdoctoral Researcher at Center for Astrochemical Studies (CAS) — *2025/12 – present*
Supervisors: Dr. Silvia Spezzano, Dr. Paola Caselli

Education

Leiden Observatory, Leiden University — Leiden, The Netherlands
Ph.D. student, Astronomy, Astrochemistry — *2021/10 – 2025/10*
Promoters: Prof. Ewine van Dishoeck, Prof. Harold Linnartz†, Dr. Martijn van Gelder
- Thesis: *A song of gas and ice: the formation and evolution of complex organic molecules*

Leiden Observatory, Leiden University — Leiden, The Netherlands
M.S. with cum laude, Astronomy, Astronomy and Data Science — *2019/09 – 2021/07*
Supervisors: Prof. Ewine van Dishoeck (1st-year), Prof. Yamila Miguel (2nd-year)
- Thesis (1st-year): *Carbon chain molecules and complex organic molecules in BHR 71*
- Thesis (2nd-year): *Observability of S and P chemistry in atmospheres of hot Jupiters*

Peking University — Beijing, China
B.S., Astronomy, Astrophysics — *2015/09 – 2019/07*
Supervisors: Prof. Gregory Herczeg, Prof. Edwin Bergin, Dr. (now Prof.) Ke Zhang
- Thesis (main body in English): *Using HD and CO isotopologues to probe the gas mass in protoplanetary disks*

University of Michigan, Ann Arbor — Ann Arbor, the United States
Undergraduate exchange program — *2018/09/12 – 2019/01/31*

Max Planck Institute for Astronomy (MPIA) — Heidelberg, Germany
Summer intern for mini research project — *2018/07/09 – 2018/08/31*

Approved Observing Proposals (selected)

As PI	JWST Cycle 4	44.3 h ($273,314)	GO 8887
As PI	ALMA Cycle 12	52.8 h (rank B)	2025.1.01411.S
As PI	ALMA Cycle 12	13.2 h (rank B)	2025.1.01654.S
As PI	ALMA Cycle 11	13 h (rank B)	2024.1.01654.S
As co-I	SMA 2024A	12.8 h (rank B)	2024A-S019

Publications (h-index: 10)

NASA ADS/SciX, as of Oct 2026

As first-author: 96 total citations

[1] **Chen, Y.**, van Dishoeck, E. F., Slavicinska, K., et al. 2026, ApJ, 1005, L30
Probing the Ubiquity of Complex Ices in Protostars with JWST: The First Systematic Quantification of Weak Ice Bands between 6.8 and 7.9 μm

[2] **Chen, Y.**, Garrod, R. T., Rachid, M., et al. 2025, A&A, 696, A198 (citations: 9)
CoCCoA: Complex Chemistry in hot Cores with ALMA. The chemical evolution of acetone from ice to gas

[3] **Chen, Y.**, Rocha, W. R. M., van Dishoeck, E. F., et al. 2024, A&A, 690, A205 (citations: 43)
JOYS+: The link between the ice and gas of complex organic molecules: Comparing JWST and ALMA data of two low-mass protostars

[4] **Chen, Y.**, van Gelder, M. L., Nazari, P., et al. 2023, A&A, 678, A137 (citations: 44)
CoCCoA: Complex Chemistry in hot Cores with ALMA. Selected oxygen-bearing species

As co-author: 14 papers, 485 total citations (see complete list of publications and Google Scholar)

Conferences & Presentations

Conferences, Workshops & Schools

2026/08	**contributed talk**	The 46th COSPAR Scientific Assembly	Florence, IT
2026/06	**contributed talk**	Perspectives of Planet and Star Formation	Heidelberg, DE
2025/10	**contributed talk**	Perspective in Star Formation: senior insights, young visions	Shanghai, CN
2025/08	**poster**	Lorentz Workshop: Interstellar disparity: N vs O-bearing molecules	Leiden, NL
2025/05	**invited talk**	Center for Interstellar Catalysis (InterCat) annual retreat	Aarhus, DK
2025/03	**contributed talk**	ESO Workshop: Towards new frontiers: the astrochemical journey from young stellar nurseries to exoplanets	Garching, DE
2025/02	**attendance**	Global Astrochemistry Lecture Series (GALS) 2025	online
2024/07	**contributed talk**	EAS Annual Meeting 2024	Padova, IT
2024/06	**attendance**	Exoplanet 5	Leiden, NL
2024/06	**invited talk**	InterCat annual retreat	Iceland
2024/01	**contributed talk**	ALMA & JWST Joint Science Day	Groningen, NL
2023/12	**contributed talk**	Exoplanets & Planet Formation Workshop 2023	Beijing, CN
2023/11	**contributed talk**	NOVA Network 2 (formation and evolution of stars and planetary systems) meeting	Leiden, NL
2023/07	**poster**	Kavli–IAU Symposium Astrochemistry VIII: From the First Galaxies to the Formation of Habitable Worlds	Traverse City, US
2023/06	**invited talk**	InterCat annual retreat (international)	Aarhus, DK
2022/10	**invited talk**	InterCat annual retreat	Copenhagen, DK
2022/10	**attendance**	Niels Bohr Legacy Symposium in Astrochemistry	Copenhagen, DK
2022/08	**attendance**	IMPRS Summer School: Astronomy, astrochemistry & the origin of life	Heidelberg, DE
2020/07	**e-poster**	EAS Annual Meeting 2020	Leiden, NL
2019/01	**e-poster**	The 233rd AAS Meeting	Seattle, US

Talks during research visits

2025/11	**seminar talk**	Sun Yet-Sen University	Zhuhai, CN
2025/10	**seminar talk**	Kavli Institute for Astronomy and Astrophysics (KIAA), Peking University	Beijing, CN
2024/01	**seminar talk**	Shanghai Astronomical Observatory	Shanghai, CN
2023/12	**seminar talk**	Guangzhou University	Guangzhou, CN
2023/07	**workshop talk**	University of Michigan, Ann Arbor	Ann Arbor, US

Teaching & Mentoring & Outreach

Since 2026/06	Non-scientific mentor of a first-year PhD student at MPE
2022, 2023, 2024, 2025 spring	Teaching assistant of the compulsory master course *Stellar Structure and Evolution* for four semesters: giving biweekly tutorial classes (incl. hands-on sessions of MESA simulations), grading essays and exams, invigilating exams, answering emails
2024/02–12	Supervision of the 2nd-year Master's research project at Leiden Observatory: *Unveiling sulphur's role in protostellar chemistry: A comparative study of gas and ice in BHR 71*
2022/02–12	Supervision of the 1st-year Master's research project at Leiden Observatory: *What sets the abundances of complex organic molecules?*
2022/07/13, 14	JWST 1:10 model demonstration at the EuroScience Open Forum (ESOF) 2022, Leiden
2022/05/22	JWST 1:10 model demonstration on the Faculty of Science Family Day, Leiden University
2018/08/13	Sole organizer and guide of a successful amateur observation of the Perseids with around 30 participants at the suburb of Heidelberg
2016/01, 2018/01, 2021/02, 2024/01	Voluntary astronomy teaching and career guidance at Guangdong Experimental High School (all on site except the one in 2021)
Since 2015/09	Permanent senior member of the Youth Astronomy Society at Peking University
2013/05 – 2014/05	Vice president of the Astronomy Association at Guangdong Experimental High School: organizing and teaching astronomy classes and amateur observation activities

Honors & Awards

2023/10	The LKBF (Leids Kerkhoven-Bosscha Fonds) subsidies for conference attendance (€900)
2018/09	Scholarship for excellent undergraduates studying abroad, China Scholarship Council ($4800)
2017 & 2018	Scholarship of Chinese National Astronomy Observatories, Chinese Academy of Sciences
2013 & 2014	Silver Prize, Chinese National Astronomy Olympiad (for high school students)

Complete List of Publications

Hyperlinks have been attached to the titles.

As first-author: 96 total citations

[1] *Probing the Ubiquity of Complex Ices in Protostars with JWST: The First Systematic Quantification of Weak Ice Bands between 6.8 and 7.9 μm.* ★
Chen, Y., van Dishoeck, E. F., Slavicinska, K., Tychoniec, Ł., van Gelder, M. L., Francis, L., Boogert, A. C. A., Narang, M., Schravendijk, C. A. F., Beuther, H., Caratti o Garatti, A., Eenhorst, E. K. H., Gieser, C., Henning, T., Güdel, M., Le Gouellec, V. J. M., McClure, M. K., Nazari, P., Reyes, S. D., Tolman, M., and Yang, Y.-L., 2026, The Astrophysical Journal Letters, 1005, L30.

[2] *CoCCoA: Complex Chemistry in hot Cores with ALMA. The chemical evolution of acetone from ice to gas.*
Chen, Y., Garrod, R. T., Rachid, M., van Dishoeck, E. F., Brogan, C., Loomis, R., Lipnicky, A., and McGuire, B. A., 2025, Astronomy & Astrophysics, 696, A198 (citations: 9).

[3] *JOYS+: The link between the ice and gas of complex organic molecules: Comparing JWST and ALMA data of two low-mass protostars.* ★
Chen, Y., Rocha, W. R. M., van Dishoeck, E. F., van Gelder, M. L., Nazari, P., Slavicinska, K., Francis, L., Tabone, B., Ressler, M. E., Klaassen, P. D., Beuther, H., Boogert, A. C. A., Gieser, C., Kavanagh, P. J., Perotti, G., Le Gouellec, V. J. M., Majumdar, L., Güdel, M., and Henning, T., 2024, Astronomy & Astrophysics, 690, A205 (citations: 43).

[4] *CoCCoA: Complex Chemistry in hot Cores with ALMA. Selected oxygen-bearing species.*
Chen, Y., van Gelder, M. L., Nazari, P., Brogan, C. L., van Dishoeck, E. F., Linnartz, H., Jørgensen, J. K., Hunter, T. R., Wilkins, O. H., Blake, G. A., Caselli, P., Chuang, K.-J., Codella, C., Cooke, I., Drozdovskaya, M. N., Garrod, R. T., Ioppolo, S., Jin, M., Kulterer, B. M., Ligterink, N. F. W., Lipnicky, A., Loomis, R., Rachid, M. G., Spezzano, S., and McGuire, B. A., 2023, Astronomy & Astrophysics, 678, A137 (citations: 44).

As co-author: 485 total citations

[1] *JOYS+: Analyses of OCN^-, N_2O, NO, and complex cyanides in ices: Thermal processing results in a modest increase of OCN^- ice.*
Nazari, P., Brunken, N., **Chen, Y.**, Slavicinska, K., van Dishoeck, E. F., Rocha, W. R. M., Boogert, A. C. A., Navarro, M. G., Le Gouellec, V. J. M., Francis, L., Tychoniec, Ł., Caratti o Garatti, A., Gieser, C., Greene, T. P., and Kavanagh, P. J., 2026, Astronomy & Astrophysics, 711, A240.

[2] *JOYS: Launch and destruction of dust in protostellar jets: The case of BHR71-IRS1 with JWST MIRI.*
Tychoniec, Ł., Francis, L., Navarro, M. G., Vorster, J. M., van Dishoeck, E. F., Caratti o Garatti, A., Assani, K., Le Gouellec, V. J. M., Tabone, B., Klaassen, P., Janssen, A. G. M., Justtanont, K., Harsono, D., Nazari, P., Reyes, S., Slavicinska, K., Gieser, C., Bourke, T., Yang, Y.-L., Nisini, B., Giannini, T., Beuther, H., Devaraj, R., Ray, T. P., Brunken, N. G. C., **Chen, Y.**, and van Gelder, M. L., 2026, Astronomy & Astrophysics, 710, A370 (citations: 2).

[3] *JOYS: Linking the molecular ice and gas phase composition toward the high-mass hot core IRAS 18089–1732.*
Gieser, C., Rocha, W. R. M., **Chen, Y.**, Slavicinska, K., van Dishoeck, E. F., Nazari, P., Brunken, N. G. C., Francis, L., Beuther, H., Reyes-Reyes, S., Caratti o Garatti, A., Klaassen, P. D., Vorster, J. M., and Navarro, M. G., 2026, Astronomy & Astrophysics, 709, A235 (citations: 2).

[4] *JOYS: Unlocking accretion-rate diagnostics for high-mass protostars using JWST/MIRI HI lines.*
Reyes-Reyes, S. D., Beuther, H., van Dishoeck, E. F., Gieser, C., Caratti o Garatti, A., Tychoniec, Ł., Kavanagh, P. J., Klaassen, P. D., Justtanont, K., Francis, L., Le Gouellec, V. J. M., Devaraj, R., Ray, T. P., **Chen, Y.**, Navarro, M. G., Rocha, W. R. M., and van Gelder, M. L., 2026, Astronomy & Astrophysics, 709, A13 (citations: 1).

[5] *Expanding the Ice Inventory of NGC1333 IRAS 2A with INDRA Using JWST Observations: Tracing Organic Refractories and Beyond.*
Rayalacheruvu, P., Majumdar, L., Rocha, W. R. M., Ressler, M. E., Giri, P. R., Maitrey, S., Willacy, K., Lis, D. C., **Chen, Y.**, and Klaassen, P. D., 2025, The Astrophysical Journal Supplement Series, 281, 51 (citations: 10).

[6] *Sulfur oxides tracing streamers and shocks at low-mass protostellar disk–envelope interfaces.*
Liu, X.-C., van Dishoeck, E. F., Hogerheijde, M. R., van Gelder, M. L., **Chen, Y.**, Liu, T., van't Hoff, M., Drozdovskaya, M. N., Artur de la Villarmois, E., Mai, X.-F., and Tychoniec, Ł., 2025, Astronomy & Astrophysics, 701, A141 (citations: 7).

[7] *HDO Ice Detected toward an Isolated Low-mass Protostar with JWST.*
Slavicinska, K., Tychoniec, Ł., Navarro, M. G., van Dishoeck, E. F., Tobin, J. J., van Gelder, M. L., **Chen, Y.**, Boogert, A. C. A., Drechsler, W. B., Beuther, H., Caratti o Garatti, A., Megeath, S. T., Klaassen, P. D., Looney, L. W., Kavanagh, P. J., Brunken, N. G. C., Sheehan, P., and Fischer, W. J., 2025, The Astrophysical Journal Letters, 986, L19 (citations: 17).

[8] *JWST Observations of Young protoStars (JOYS): Overview of program and early results.*
van Dishoeck, E. F., Tychoniec, Ł., Rocha, W. R. M., Slavicinska, K., Francis, L., van Gelder, M. L., Ray, T. P., Beuther, H., Caratti o Garatti, A., Brunken, N. G. C., **Chen, Y.**, Devaraj, R., Geers, V. C., Gieser, C., Greene, T. P., Justtanont, K., Le Gouellec, V. J. M., Kavanagh, P. J., Klaassen, P. D., Janssen, A. G. M., Navarro, M. G., Nazari, P., Notsu, S., Perotti, G., Ressler, M. E., Reyes, S. D., Sellek, A. D., Tabone, B., Tap, C., Theijssen, N. C. M. A., Colina, L., Güdel, M., Henning, T., Lagage, P.-O., Östlin, G., Vandenbussche, B., and Wright, G. S., 2025, Astronomy & Astrophysics, 699, A361 (citations: 47).

[9] *JWST Observations of Young protoStars (JOYS): Overview of gaseous molecular emission and absorption in low-mass protostars.*
van Gelder, M. L., Francis, L., van Dishoeck, E. F., Tychoniec, Ł., Ray, T. P., Beuther, H., Caratti o Garatti, A., **Chen, Y.**, Devaraj, R., Gieser, C., Justtanont, K., Kavanagh, P. J., Nazari, P., Reyes, S., Rocha, W. R. M., Slavicinska, K., Güdel, M., Henning, T., Lagage, P.-O., and Wright, G., 2024, Astronomy & Astrophysics, 692, A197 (citations: 33).

[10] *JWST detections of amorphous and crystalline HDO ice toward massive protostars.*
Slavicinska, K., van Dishoeck, E. F., Tychoniec, Ł., Nazari, P., Rubinstein, A. E., Gutermuth, R., Tyagi, H., **Chen, Y.**, Brunken, N. G. C., Rocha, W. R. M., Manoj, P., Narang, M., Thomas Megeath, S., Yang, Y.-L., Looney, L. W., Tobin, J. J., Beuther, H., Bourke, T. L., Linnartz, H., Federman, S., Watson, D. M., and Linz, H., 2024, Astronomy & Astrophysics, 688, A29 (citations: 35).

[11] *JWST Observations of Young protoStars (JOYS): Linked accretion and ejection in a Class I protobinary system.*
Tychoniec, Ł., van Gelder, M. L., van Dishoeck, E. F., Francis, L., Rocha, W. R. M., Caratti o Garatti, A., Beuther, H., Gieser, C., Justtanont, K., Linnartz, H., Le Gouellec, V. J. M., Perotti, G., Devaraj, R., Tabone, B., Ray, T. P., Brunken, N. G. C., **Chen, Y.**, Kavanagh, P. J., Klaassen, P. D., Slavicinska, K., Güdel, M., and Östlin, G., 2024, Astronomy & Astrophysics, 687, A36 (citations: 44).

[12] *JWST Observations of Young protoStars (JOYS+): Detecting icy complex organic molecules and ions. I. CH_4, SO_2, $HCOO^-$, OCN^-, H_2CO, HCOOH, CH_3CH_2OH, CH_3CHO, CH_3OCHO, and CH_3COOH.*
Rocha, W. R. M., van Dishoeck, E. F., Ressler, M. E., van Gelder, M. L., Slavicinska, K., Brunken, N. G. C., Linnartz, H., Ray, T. P., Beuther, H., Caratti o Garatti, A., Geers, V., Kavanagh, P. J., Klaassen, P. D., Justtanont, K., **Chen, Y.**, Francis, L., Gieser, C., Perotti, G., Tychoniec, Ł., Barsony, M., Majumdar, L., Le Gouellec, V. J. M., Chu, L. E. U., Lew, B. W. P., Henning, T., and Wright, G., 2024, Astronomy & Astrophysics, 683, A124 (citations: 138).

[13] *JOYS+: Mid-infrared detection of gas-phase SO_2 emission in a low-mass protostar. The case of NGC 1333 IRAS 2A: Hot core or accretion shock?*
van Gelder, M. L., Ressler, M. E., van Dishoeck, E. F., Nazari, P., Tabone, B., Black, J. H., Tychoniec, Ł., Francis, L., Barsony, M., Beuther, H., Caratti o Garatti, A., **Chen, Y.**, Gieser, C., Le Gouellec, V. J. M., Kavanagh, P. J., Klaassen, P. D., Lew, B. W. P., Linnartz, H., Majumdar, L., Perotti, G., and Rocha, W. R. M., 2024, Astronomy & Astrophysics, 682, A78 (citations: 31).

[14] *Which molecule traces what: Chemical diagnostics of protostellar sources.*
Tychoniec, Ł., van Dishoeck, E. F., van't Hoff, M. L. R., van Gelder, M. L., Tabone, B., **Chen, Y.**, Harsono, D., Hull, C. L. H., Hogerheijde, M. R., Murillo, N. M., and Tobin, J. J., 2021, Astronomy & Astrophysics, 655, A65 (citations: 118).